DO NOW — RECALL

BIOS, drivers and virtual machines

Name: _____ Class: _____ Date: _____

1. What does an OS driver do?

2. Define multi-user OS.

3. What executes before the OS is loaded?

4. Simplify (¬A∧B)∨(A∧B). Show factoring.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Interfaces with specific hardware.

2. Model answer

Supports separate user sessions and permissions.

3. Model answer

Platform firmware and boot-loading code.

4. Model answer

B∧(¬A∨A)=B∧1=B.

Vocabulary — know these terms

UEFI — Modern system boot firmware.

driver — Hardware-specific software interface.

VM — Virtual machine.

hypervisor — VM hosting software.

NEW LEARNING

Read, annotate and explain

Understand

BIOS/UEFI firmware initialises key hardware and starts the boot process.

Trace

Device drivers connect OS abstractions to the instructions and data expected by actual peripherals.

Evaluate

A hypervisor provides virtual machines for isolation, testing and consolidation, though CPU/RAM/storage overhead remains.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A printer is connected but software cannot print; name an OS dependency.

Predict first

A missing or incompatible driver can prevent correct communication with a device.

Model and check

Explain why each value follows from the earlier step.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State roles of UEFI/BIOS, boot loader, driver and hypervisor. [4]

Question 2

Explain how a compatible printer driver enables printing. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate guest virtual machines for school testing of unfamiliar operating systems. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A virtual machine has its own wholly separate physical processor.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.